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Faculty of Electrical Engineering, Banja Luka
Department of Computing and Informatics

Internet safety

Project assignment

Task

Realize the information system of the insurance company, which consists of two applications: *administrator* and *client.* The system allows clients to view, purchase and manage insurance policies (life, travel, property insurance, etc.). The administrator application enables the management of user accounts and services offered to clients.

A new client registers on the system and, after registration, can log on to the system using two-factor authentication. In the first step, the client enters a username and password, after which he receives a verification code via *e-mail,* which must be entered in the next step of registration.

It is necessary to implement a Single Sign On (SSO) mechanism, so that users (administrators, employees), who have access to both applications, can authenticate in either application and have access to the entire system.

Clients can purchase a specific policy, and for the purposes of payment simulation, it is necessary to integrate one of the existing payment systems *(Stripe, PayPal sanbox* etc.). Payment is made exclusively in test mode, without generating actual charges, and transaction data must be saved in the database in an appropriate manner. Upon successful purchase, the client will receive the purchased policy in PDF format via email.

Figure 1 shows an example of the system architecture. *Access Controller* is a component through which users interact with the system (user authentication and authorization, session monitoring, detection of security-sensitive requests, etc.). If it detects a potentially malicious request (for example, entering a significantly larger payment amount, etc.), *the Access Controller* can automatically close the session with the user. The SIEM component is responsible for monitoring and logging all security-sensitive requests. *Nagios* needs to be configured for monitoring all key services.

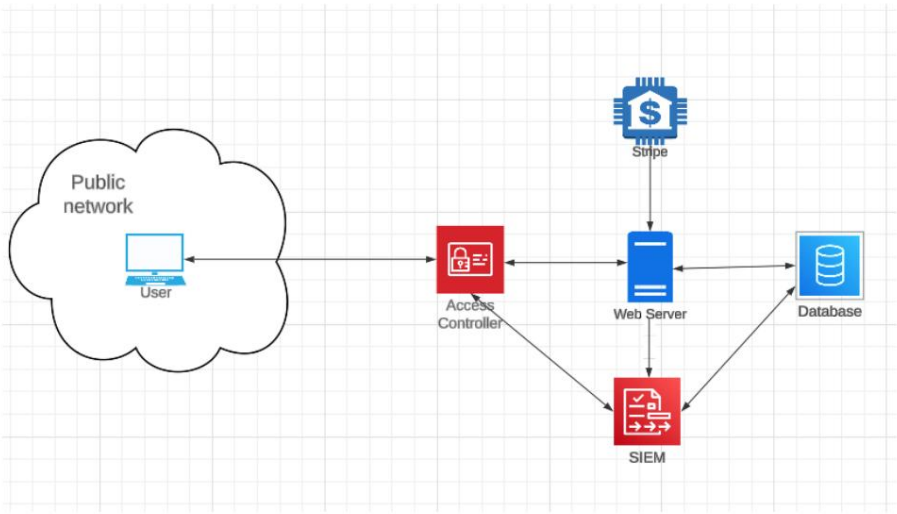


Figure 1 – System architecture

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All the details of the task that are not precisely specified should be realized in an arbitrary way. Implement user interface details in an arbitrary way. The system should be *web*- based. The use of arbitrary programming language is allowed, as well as arbitrary technologies necessary for the realization of technical details.

Students who successfully defend the project assignment gain the right to take the oral part of the exam. Before the defense, it is necessary to upload the complete source code of the project assignment to *moodle.* The assignment is valid from the first term of the January-February exam period in 2025 and is valid until the publication of the next assignment.